

Dialogue systems and question answering - introduction

today: dialogue systems

Dag Wedelin, 2020

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Partly following Jurafsky and Martin

*Also inspiration from S. Nilsson
lecture in dialogue systems (GU)*

Chatbots for arbitrary conversation

“How do you feel today?”

In practice the term chatbot is used more widely

Dia logos “through language”

Digital assistants for tasks

“Where is the nearest grocery store?”

Early systems rule-based

Modern systems use a combination of rules and ML-based techniques

properties of human
conversation

A dialogue is a sequence of turns

In spoken dialogue it can be difficult to know when a turn ends

C₁: ...I need to travel in May.
A₁: And, what day in May did you want to travel?
C₂: OK uh I need to be there for a meeting that's from the 12th to the 15th.
A₂: And you're flying into what city?
C₃: Seattle.
A₃: And what time would you like to leave Pittsburgh?
C₄: Uh hmm I don't think there's many options for non-stop.
A₄: Right. There's three non-stops today.
C₅: What are they?
A₅: The first one departs PGH at 10:00am arrives Seattle at 12:05 their time. The second flight departs PGH at 5:55pm, arrives Seattle at 8pm. And the last flight departs PGH at 8:15pm arrives Seattle at 10:28pm.
C₆: OK I'll take the 5ish flight on the night before on the 11th.
A₆: On the 11th? OK. Departing at 5:55pm arrives Seattle at 8pm, U.S. Air flight 115.
C₇: OK.
A₇: And you said returning on May 15th?
C₈: Uh, yeah, at the end of the day.
A₈: OK. There's #two non-stops ...#
C₉: #Act...actually #, what day of the week is the 15th?
A₉: It's a Friday.
C₁₀: Uh hmm. I would consider staying there an extra day til Sunday.
A₁₀: OK...OK. On Sunday I have .

Speech acts

Constatives:	committing the speaker to something's being the case (<i>answering, claiming, confirming, denying, disagreeing, stating</i>)
Directives:	attempts by the speaker to get the addressee to do something (<i>advising, asking, forbidding, inviting, ordering, requesting</i>)
Commissives:	committing the speaker to some future course of action (<i>promising, planning, vowing, betting, opposing</i>)
Acknowledgments:	express the speaker's attitude regarding the hearer with respect to some social action (<i>apologizing, greeting, thanking, accepting an acknowledgment</i>)

dialogue structure
(e.g. *answer follows question*)

sub-dialogue

clarification

grounding

Initiative in
conversation

user-initiative

system-initiative

mixed initiative

Inference and implicature

A₁: And, what day in May did you want to travel?

C₂: OK uh I need to be there for a meeting that's from the 12th to the 15th.

A: I am out of gas.

B: There is a gas station round the corner. +> The gas station is open.

chatbots

ELIZA
(Weizenbaum,
1966)

Men are all alike.

IN WHAT WAY

They're always bugging us about something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time.

I AM SORRY TO HEAR YOU ARE DEPRESSED

::

WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU?

My father

YOUR FATHER

You are like my father in some ways.

WHAT RESEMBLANCE DO YOU SEE

You are not very aggressive but I think you don't want me to notice that.

WHAT MAKES YOU THINK I AM NOT AGGRESSIVE

You don't argue with me.

WHY DO YOU THINK I DON'T ARGUE WITH YOU

You are afraid of me.

DOES IT PLEASE YOU TO BELIEVE I'M AFRAID OF YOU

My father is afraid of everybody.

WHAT ELSE COMES TO MIND WHEN YOU THINK OF YOUR FATHER

Bullies.

DOES THAT HAVE ANYTHING TO DO WITH THE FACT THAT YOUR
BOYFRIEND MADE YOU COME HERE

Figure 26.4 Sample ELIZA dialogue from Weizenbaum (1966).

ELIZA
(Weizenbaum,
1966)

```
(Ø YOU Ø ME) [pattern]  
->  
(WHAT MAKES YOU THINK I 3 YOU) [transform]
```

You hate me

into:

WHAT MAKES YOU THINK I HATE YOU

Keyword based

Higher rank for specific keywords. Prefers
response based on most specific keyword

Memory based on keyword MY

PARRY (Colby
et al, 1971)

*Acts as a paranoid
person*

*Includes model of
mental state that
influences the
conversation*

interviewer: Why are you in the hospital?

parry: I shouldn't be here.

interviewer: Who brought you here?

parry: The police.

interviewer: What trouble did you have with the police?

parry: Cops don't do their jobs.

interviewer: What do they do instead?

parry: Cops arrest the wrong people.

interviewer: They arrest the right ones sometimes.

parry: The law should get those Italian crooks.

interviewer: Are you Italian?

parry: My background is British, but I was born in this country.

interviewer: Do you know any Italian crooks?

parry: I try to avoid the underworld.

interviewer: Are you afraid of the underworld?

parry: The mafia are the worst gangsters.

interviewer: Are you connected with the mafia?

parry: You know, they know me.

ALICE

(Wallace, 1995)

Human: how are you

A.L.I.C.E: I am fine, thank you.

Human: i have to work too much

A.L.I.C.E: Soon computers will make your work obsolete.

Human: then i can go sailing

A.L.I.C.E: Interesting deduction. Oh you can, can you?

*modern version of ELIZA with
programming language AIML
(AI Markup Language)*

Corpus-based chatbots

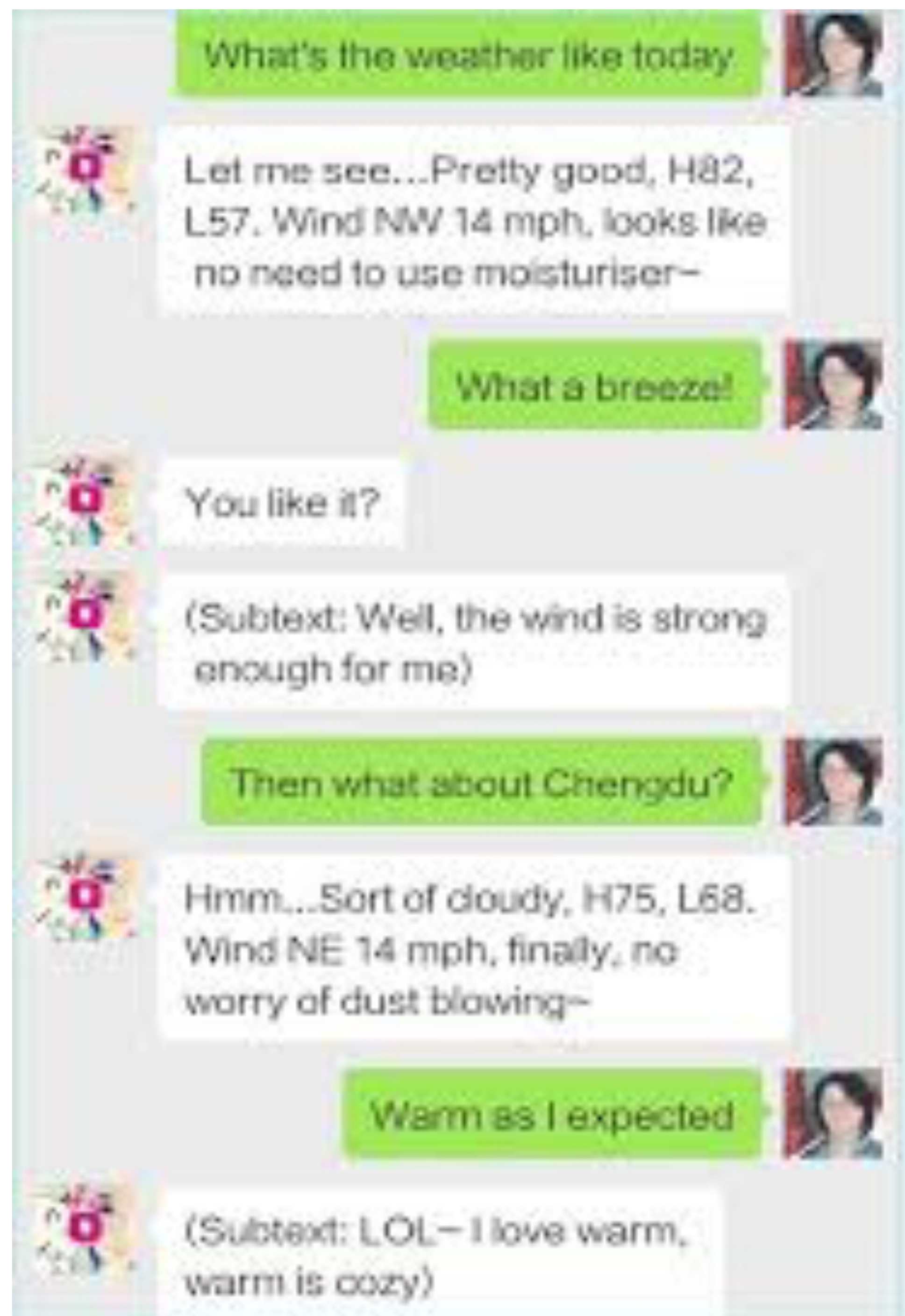
Based on very large datasets of
real conversations

Also possible with non-
dialogue corpora

Response generation: not rules
but single response based on
user's last turn (or two)

XIAOice

(popular Chinese
chatbot by Microsoft)



IR-based chatbots

1. Return response to most similar user turn
2. Return most similar response

Similarity e.g. based on word vector or word embedding

Can also use information about user or sentiment...

ML-based encoder-decoder chatbot

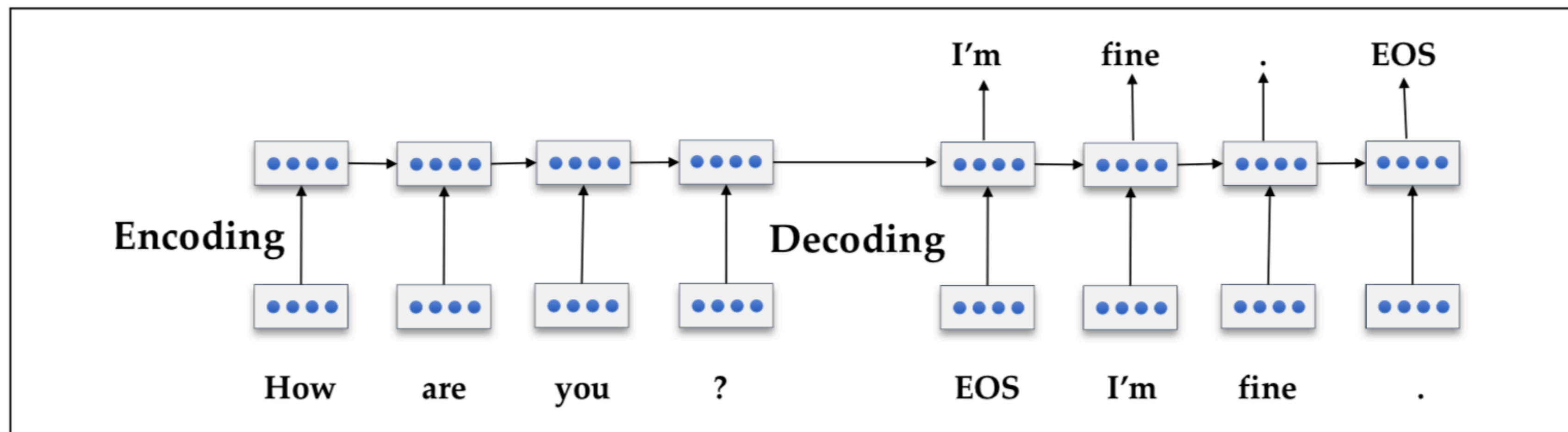


Figure 26.6 An encoder decoder model for neural response generation in dialogue.

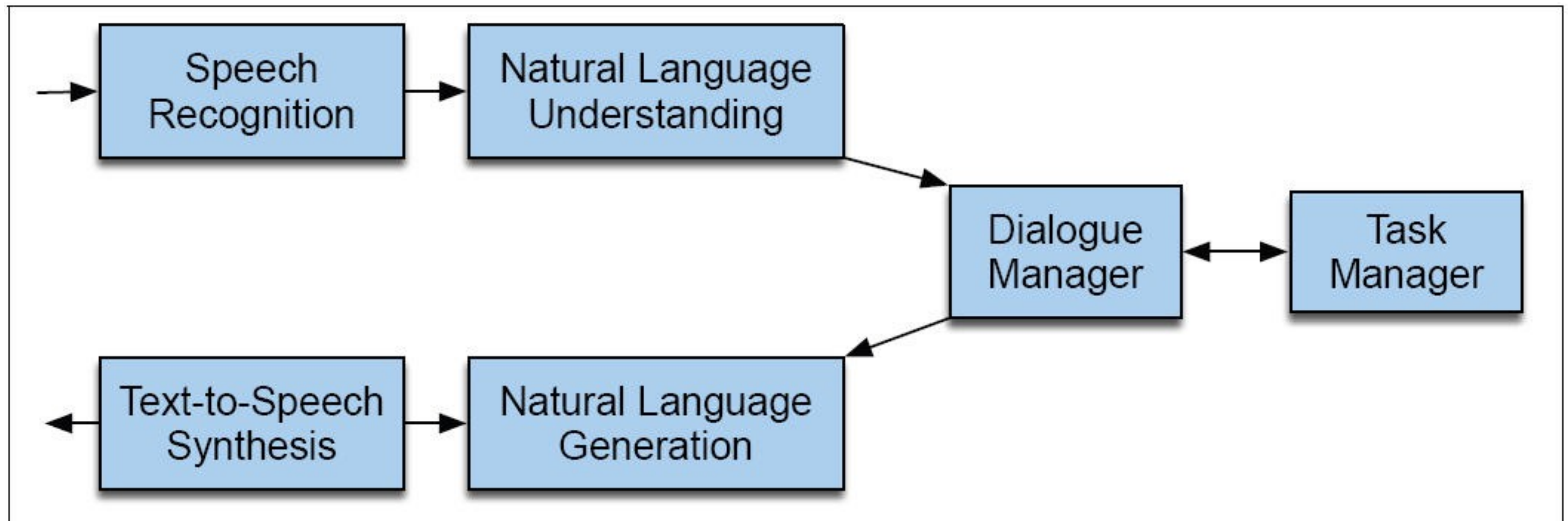
*More advanced ML-
techniques can be used...*

Microsoft TAY on Twitter

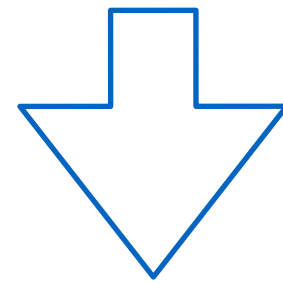


task-based dialogue systems
for digital assistants

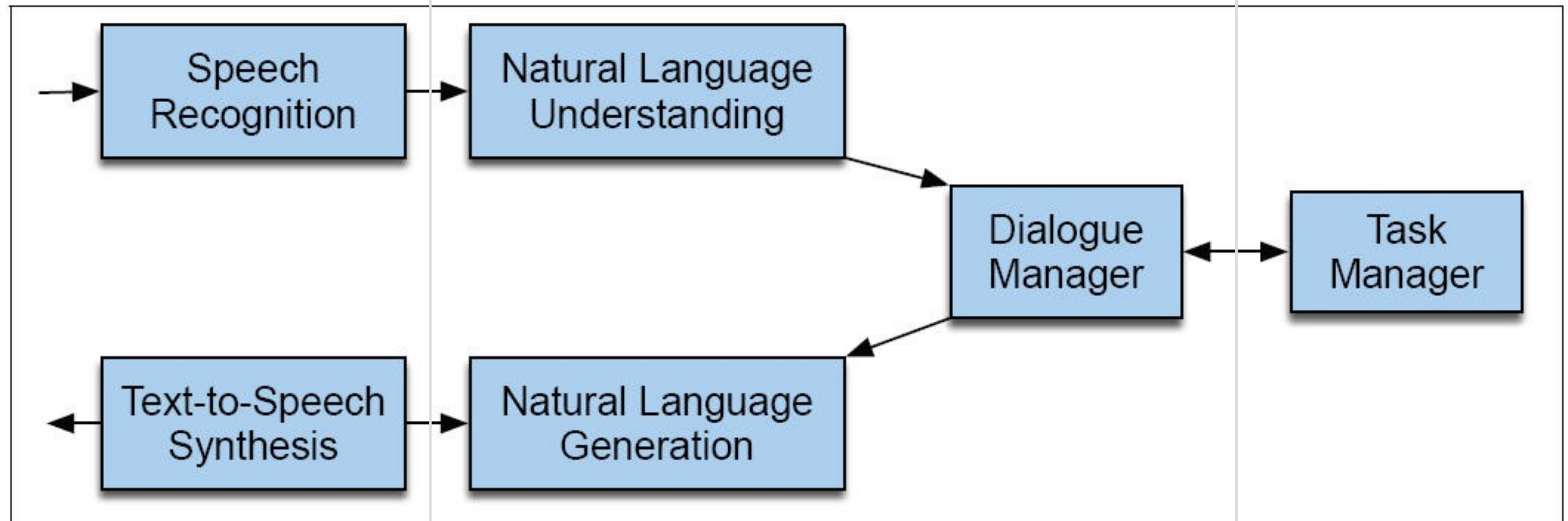
Dialogue system architecture



*statistical/NN-
based*

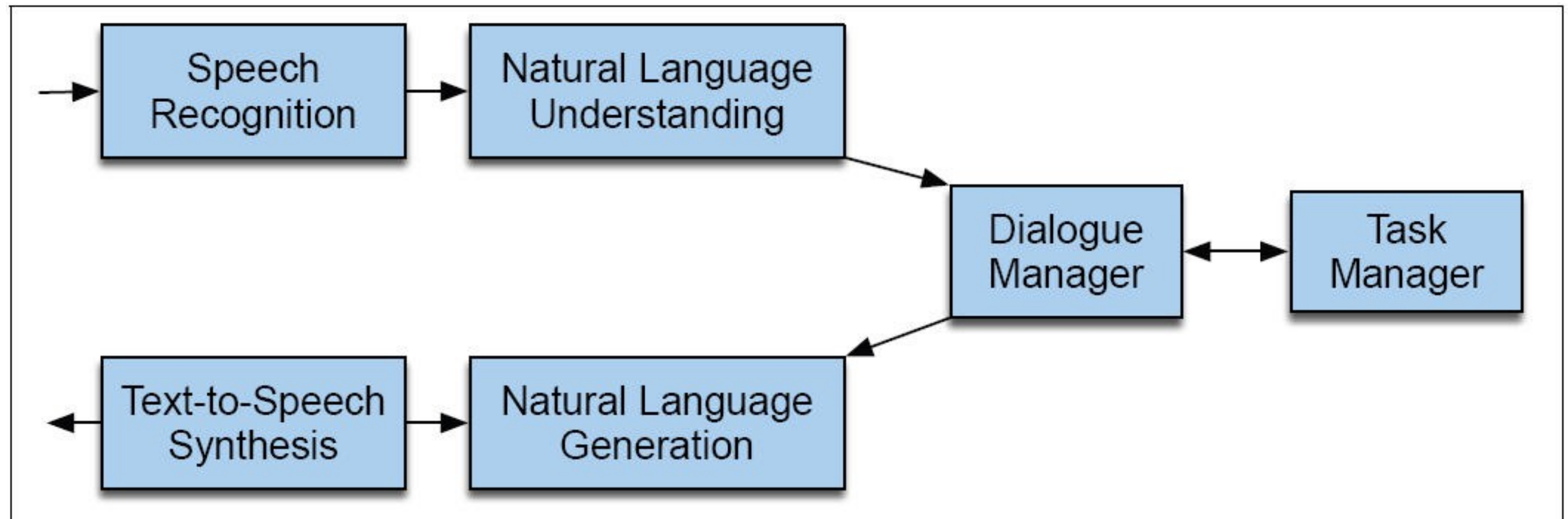
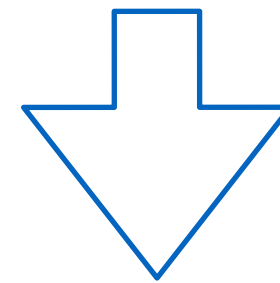


*we will mostly focus on
the text-based dialogue*

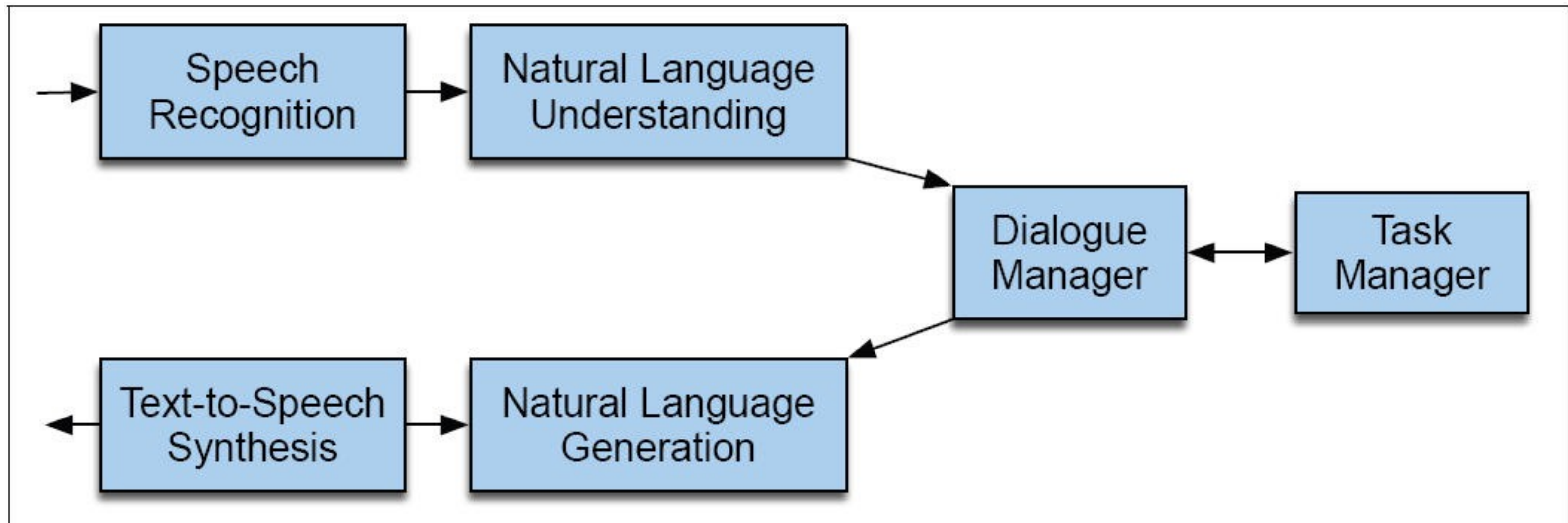
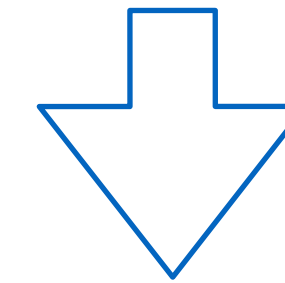


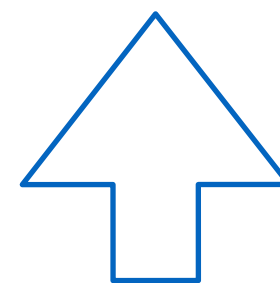
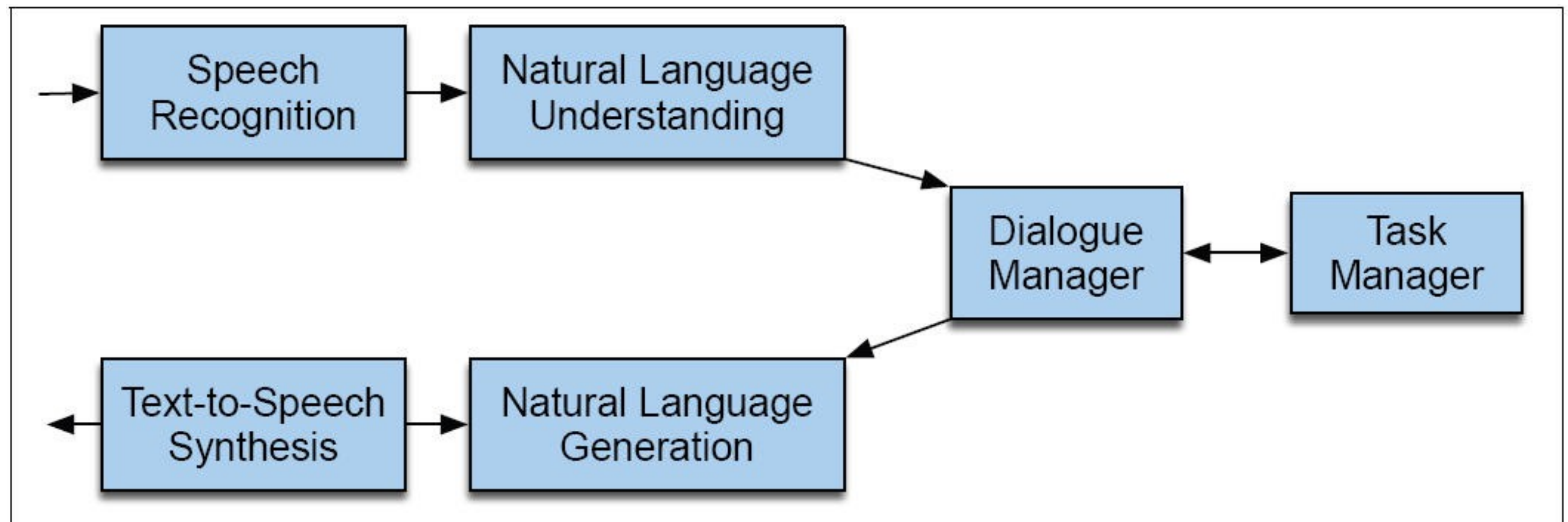
extract meaning

*rule-based or
increasingly ML-
based*



*commercial dialogue
managers are mostly
rule-based*

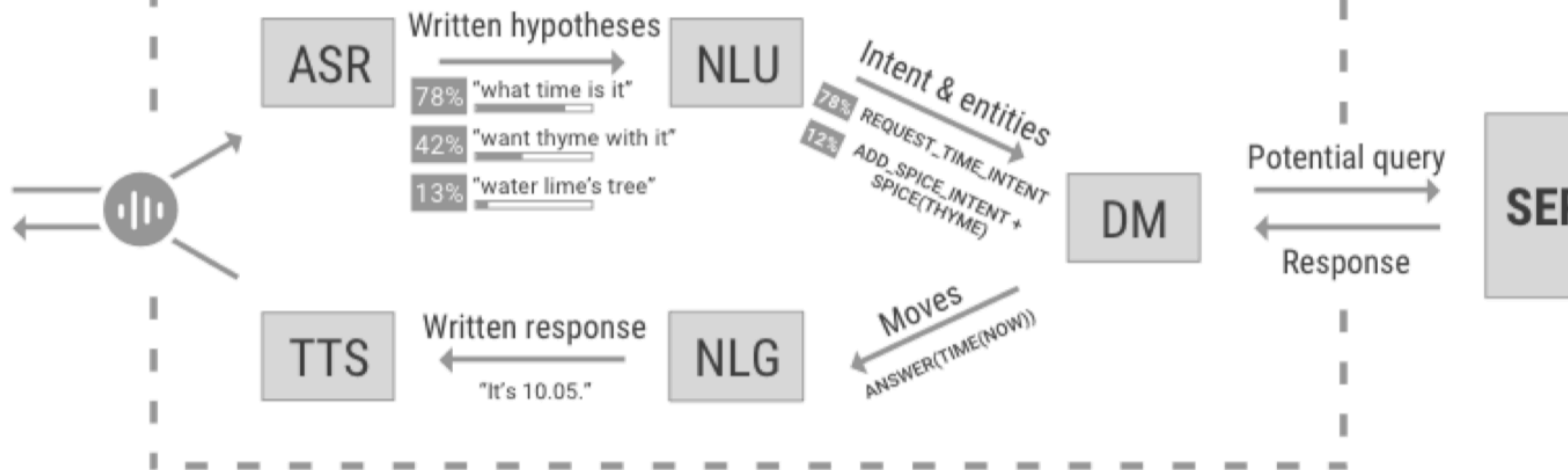




mostly template-based

*possibly learned from
delexicalized data*

Dialogue System



Natural language understanding (NLU)

classify domain

determine intent

extract relevant information

SHOW	→ show me i want can i see ...
DEPART_TIME_RANGE	→ (after around before) HOUR morning afternoon evening
HOUR	→ one two three four... twelve (AMPM)
FLIGHTS	→ (a) flight flights
AMPM	→ am pm
ORIGIN	→ from CITY
DESTINATION	→ to CITY
CITY	→ Boston San Francisco Denver Washington

Semantic grammars can be parsed by any CFG parsing algorithm (see Chapter 13), resulting in a hierarchical labeling of the input string with semantic node labels, as shown in Fig. 26.10.

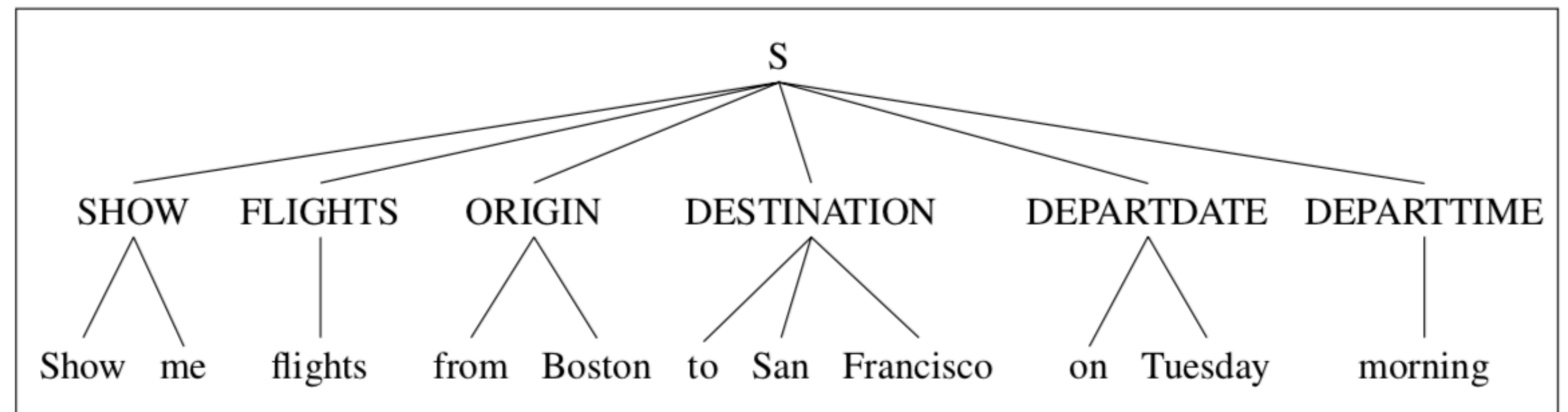
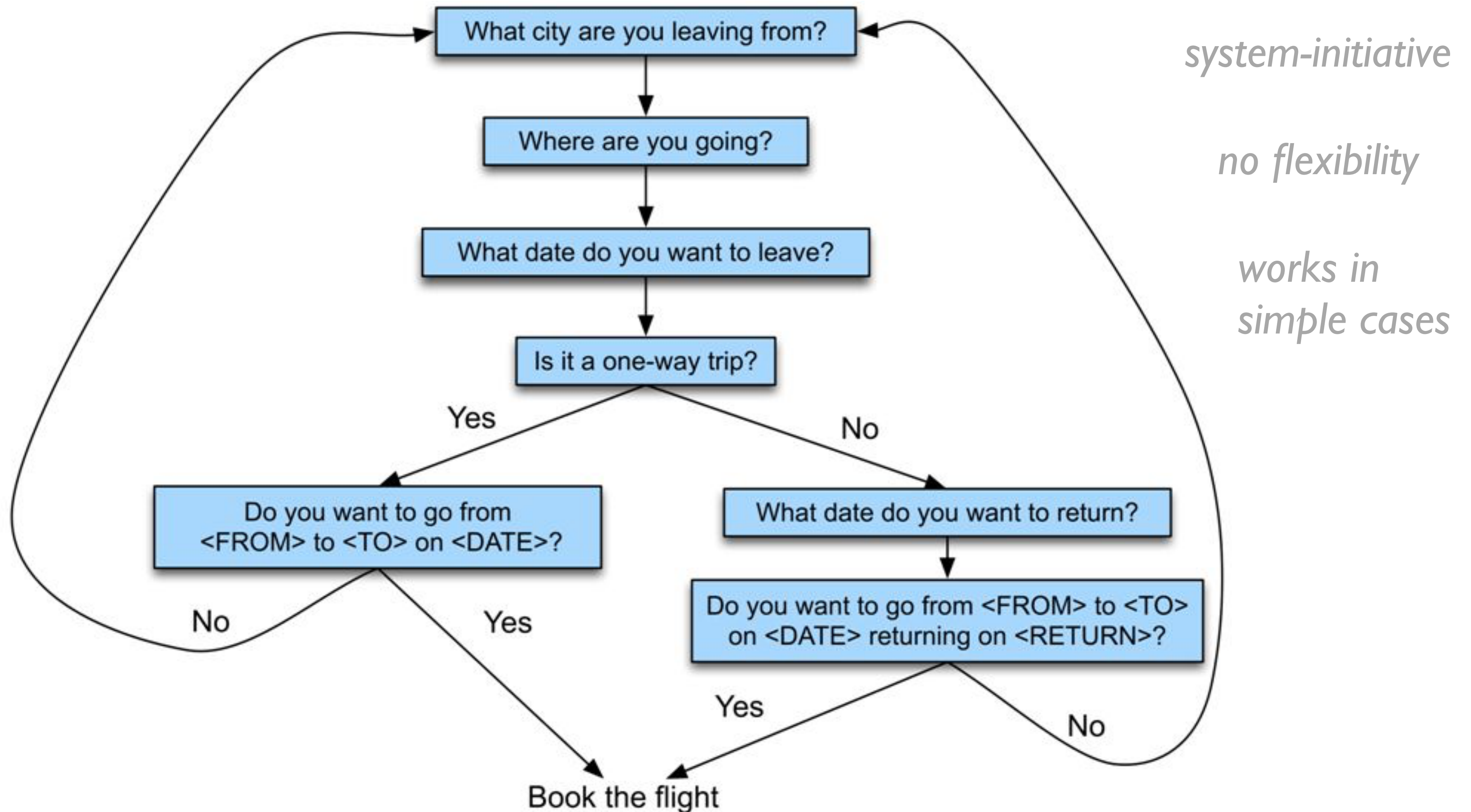


Figure 26.10 A semantic grammar parse for a user sentence, using slot names as the internal parse tree nodes.

Why dialogue management (DM)?

Otherwise the user must give all
information in one statement (no
dialogue)

Dialogue Management - finite-state based



Dialogue Management - frame-based (*or form-based*)

FRAME TRAVEL

ORIGIN CITY _____

DESTINATION CITY _____

DEPARTURE TIME _____

ARRIVAL TIME _____

Dialogue Management - frame-based

Relates to a particular common situation. Words are understood in relation to the mental “frame of reference”

FRAME TRAVEL

ORIGIN CITY _____
DESTINATION CITY _____
DEPARTURE TIME _____
ARRIVAL TIME _____

A form with empty slots constrained to certain possible values

To each slot you can associate questions (and preconditions)

Subframes

GUS (Bobrow, 1977)

All common task-based systems use frames!

Dialogue Management - frame-based

Fill all slots!

*Also ask for
confirmation, make
suggestion,...*

*Control structure for
the dialogue is based
around the frame*

FRAME TRAVEL

ORIGIN CITY _____

DESTINATION CITY _____

DEPARTURE TIME _____

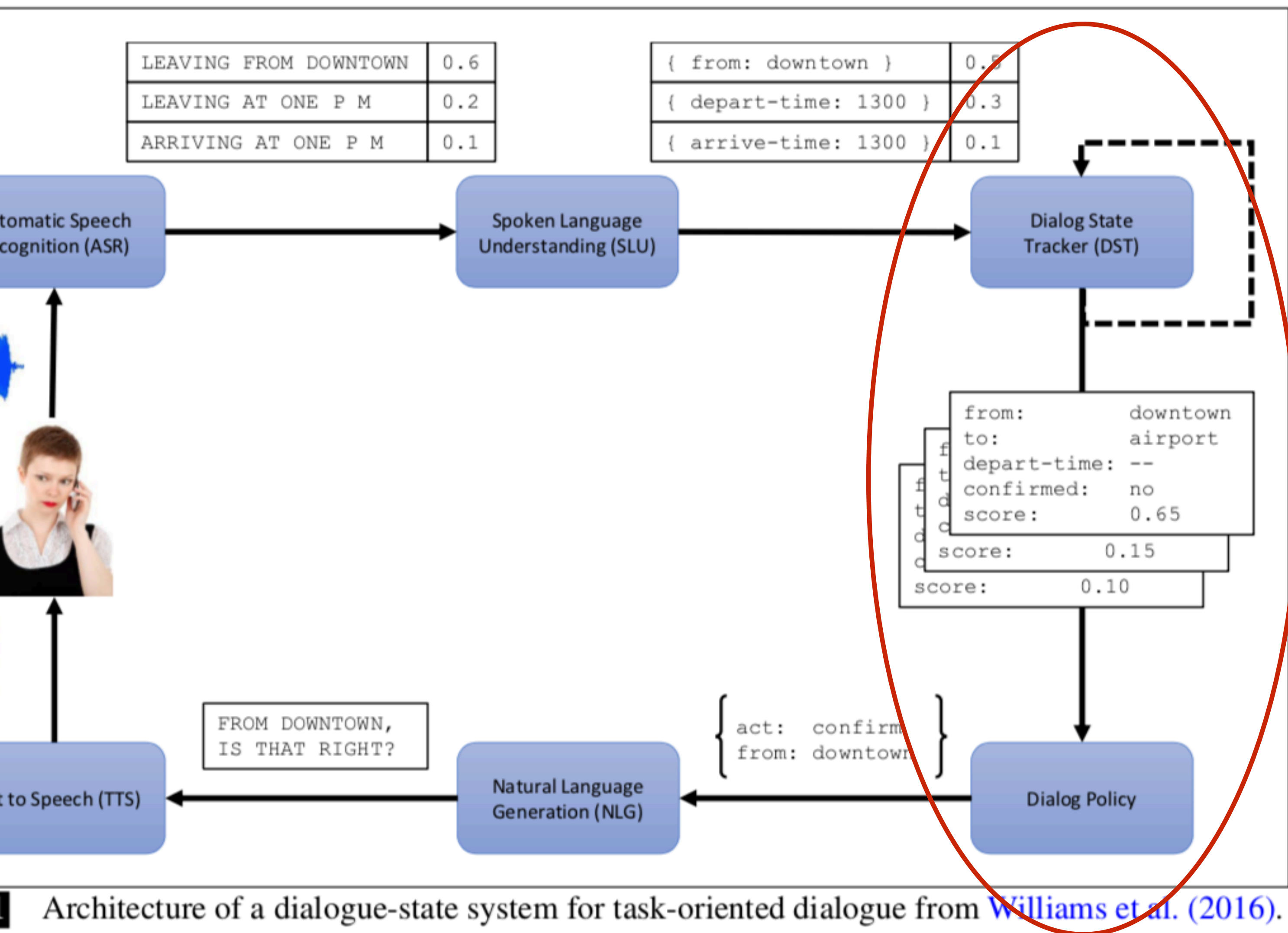
ARRIVAL TIME _____

*Multiple slots can be
filled in one turn*

*Production rules (condition->action) to
move to other frames e.g. for hotel
booking*

*See as a planning
problem?*

Dialogue Management - frame-based *with dialog state*



A rich information state beyond simple form-filling

There can be multiple frames that the system moves between

Architecture of a dialogue-state system for task-oriented dialogue from Williams et al. (2016).

Keeps different aspects separate!

Dialogue state tracking

Which slots are filled?

Most recent turn?

...

User: I'm looking for a cheaper restaurant
`inform(price=cheap)`

System: Sure. What kind - and where?

User: Thai food, somewhere downtown
`inform(price=cheap, food=Thai, area=centre)`

System: The House serves cheap Thai food

User: Where is it?
`inform(price=cheap, food=Thai, area=centre); request(address)`

System: The House is at 106 Regent Street

Dialogue policy

In principle: determine next action
based on entire previous dialogue

Simpler: current state of frame and
last turns

Includes confirming as
well as rejecting possible
misunderstandings

Can be based on rules
or be learned

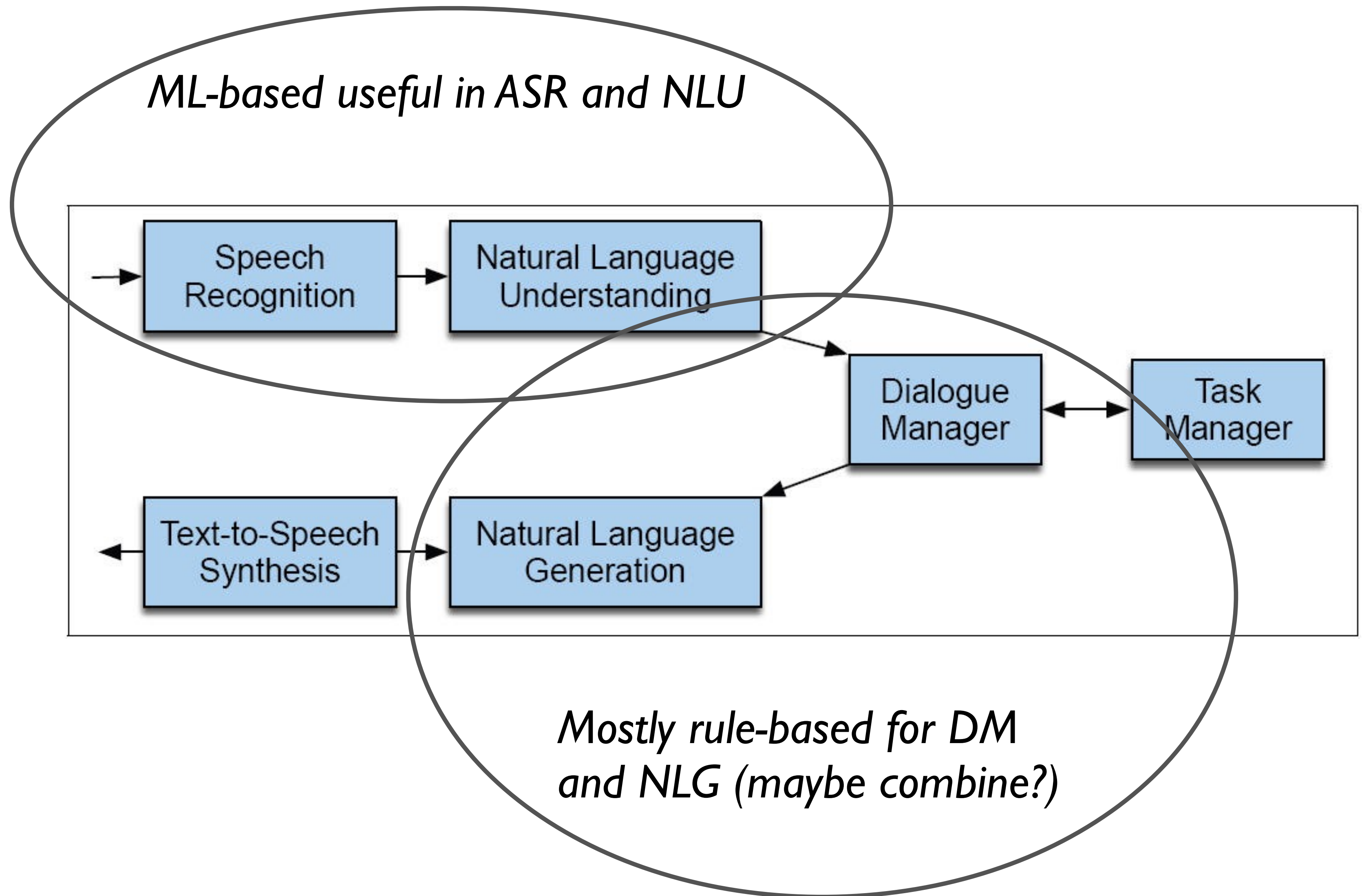
Dialogue system design

Study the user and task

Simulations and prototypes
(wizard of Oz - human
pretends to be computer)

Iterative design

Rule-based or ML-based?



To think about...

What are the limitations
of these approaches?

What is needed to
fundamentally go further?

END